

Statistical Methods for Data Science (Spring 2026)

Mini Project #3

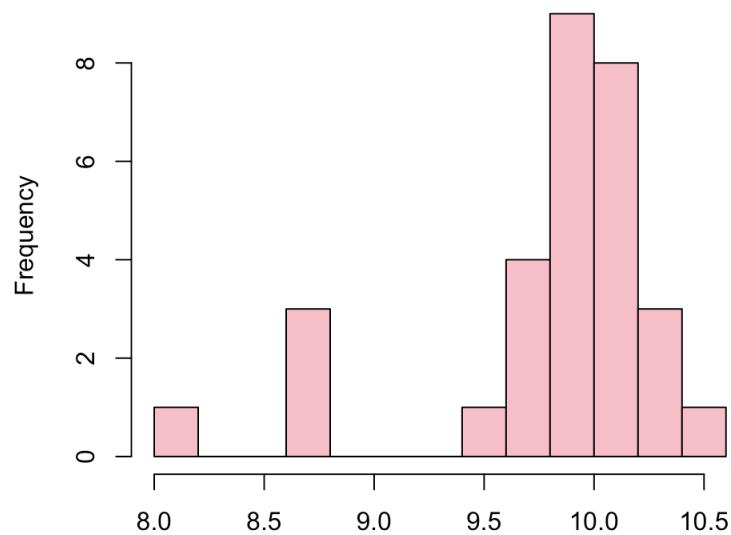
Name : FATHIMA FARIDA MOHAMMAD

SECTION 1

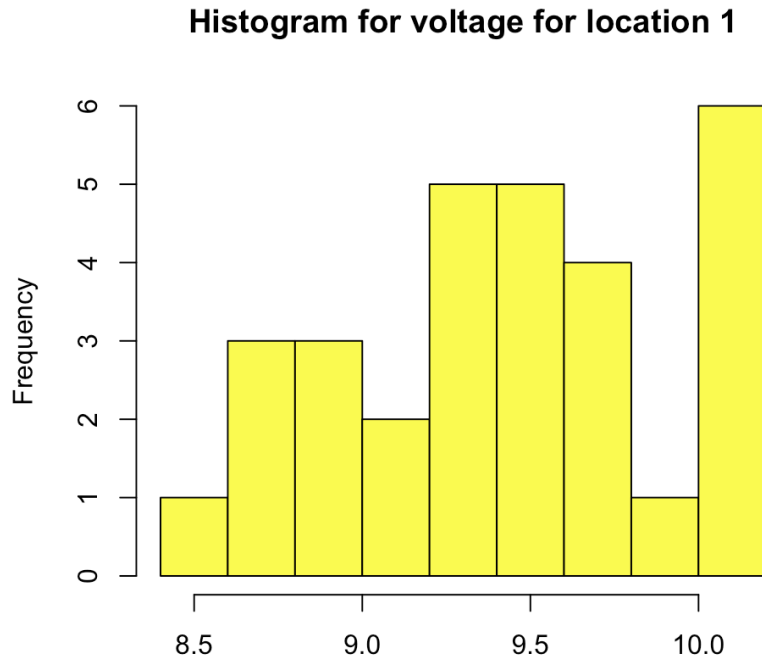
QUESTION 1

A. Exploratory Data Analysis

Histogram for voltage for location 0

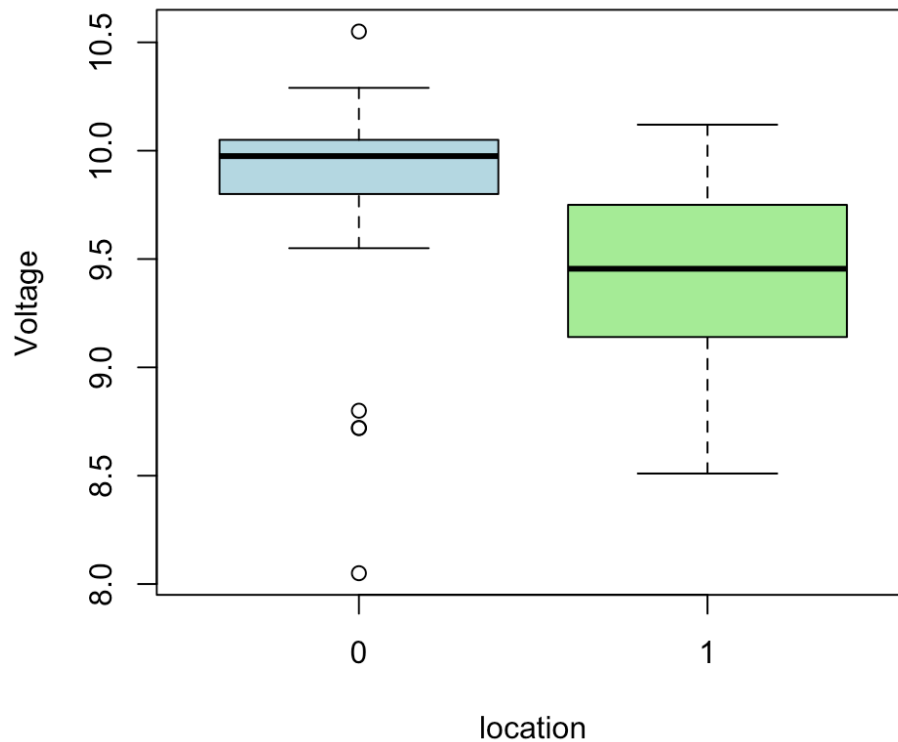


- Data from location 0 (remote) appears to be approximately normal and left-skewed.



- Location 1 data is clearly not normal and roughly symmetric. Hence we can say that the two distributions are not very similar.
- The side-by-side boxplot shows that the distribution of voltage readings differs between the two locations. The remote location (0) has a higher median voltage compared to the local location (1), indicating that, on average, voltage readings are higher at the remote site.
- Difference in spread of data: the local location exhibits a wider IQR , showing greater variability, while the remote location shows a narrower IQR but a wider overall range (8.05 to 10.55), indicating the presence of extreme values. Also, the remote location contains many outliers, particularly on the lower end.

Voltage comparision by location



Summary and stats for Remote location :

```
> summary(filter(voltage_data, location == 0)$voltage)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 8.050  9.800  9.975  9.804 10.050 10.550
> sd(filter(voltage_data, location == 0)$voltage)
[1] 0.5409155
> range(filter(voltage_data, location == 0)$voltage)
[1] 8.05 10.55
```

Summary and stats for Local location :

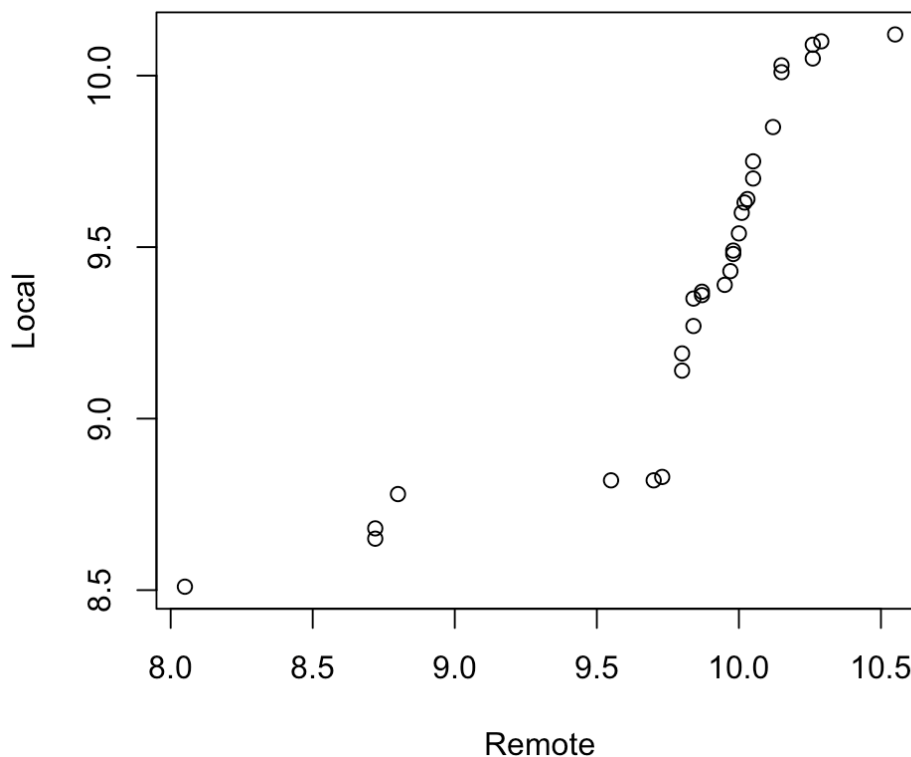
```

> summary(filter(voltage_data, location == 1)$voltage)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 8.510   9.152   9.455   9.422   9.738  10.120
> sd(filter(voltage_data, location == 1)$voltage)
[1] 0.4788757
> range(filter(voltage_data, location == 1)$voltage)
[1] 8.51 10.12

```

QQ plot

- The Q-Q plot shows that the points lie approximately along a straight line, so the two distributions have a similar overall shape. However, there are noticeable deviations from linearity, especially at the tails, indicating differences in variability and extreme values between the two groups. They are not identical.



B. Difference in population means of voltage readings at 2 locations

Welch Two Sample t-test

```
data: Local$voltage and Remote$voltage
t = -2.8911, df = 57.16, p-value = 0.005419
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.6454382 -0.1172284
sample estimates:
mean of x mean of y
 9.422333  9.803667

> cat("95% CI for (Local - Remote):\n")
95% CI for (Local - Remote):
> print(t_result$conf.int)
[1] -0.6454382 -0.1172284
attr(,"conf.level")
[1] 0.95
> cat("Mean Difference (Local - Remote):", mean(Local$voltage) - mean(Remote$voltage),
"\n")
Mean Difference (Local - Remote): -0.3813333
```

- H_0 : Mean voltage (Local) = Mean voltage (Remote)
- H_a : Mean voltages are different
- Assumption : Variances are not equal.
- 95% CI for (Local – Remote) = (-0.645, -0.117)
- The entire confidence interval is below 0, which indicates that the Local mean voltage is lower than Remote mean voltage. Since 0 is not included, there is a statistically significant difference.
- Difference of population means (Local – Remote) = -0.381
- We reject the null hypothesis with 5% confidence .
- There is statistically significant evidence of a difference, so the process cannot be assumed equivalent across locations.

C.

- The conclusion from the hypothesis test in (b) is consistent with what was suggested in the exploratory analysis (a). Both analyses indicate that the remote and local voltage distributions are not the same.

- In (a), we observed:
 - Differences in skewness (remote slightly left-skewed, local roughly symmetric but not-normal)
 - Differences in variability (local has larger IQR, remote has wider range and outliers)
- In (b), the significant confidence interval further supports that these differences are not due to random variation.
- The exploratory analysis correctly anticipated differences between locations, especially in center and spread.
- In (a), the boxplot suggested that remote has a higher median voltage than local.
- In (b), the t-test confirms this statistically:
 - Local mean = 9.422
 - Remote mean = 9.804
 - Difference is significant and negative (Local < Remote)

Section 2

A.

```

1 library(pacman)
2
3 pacman::p_load(rio)
4
5 voltage_data <- import("/Users/fathimafaridamohammad/Downloads/voltage.csv")
6
7 head(voltage_data)
8
9 ## Histogram for Location = 0
10 hist(voltage_data$voltage [voltage_data$location == '0'],
11       breaks = 10, main = "Histogram for voltage for location 0",
12       xlab = "", col="pink")
13
14 ## Histogram for Location = 1
15 hist(voltage_data$voltage [voltage_data$location == '1'],
16       breaks = 10, main = "Histogram for voltage for location 1",
17       xlab = "", col="yellow")

```

```

19 pacman::p_load(dplyr)
20
21 ##side by side boxplots for location =0 and location = 1
22 boxplot(`voltage` ~ location , data = voltage_data,
23         col= c("lightblue", "lightgreen"),
24         main = "Voltage comparision by location", ylab = "Voltage")
25
26 # Summary statistics for Location = 0
27 summary(filter(voltage_data, location == 0)$voltage)
28 sd(filter(voltage_data, location == 0)$voltage)
29 range(filter(voltage_data, location == 0)$voltage)
30
31 # Summary statistics for location = 1
32 summary(filter(voltage_data, location == 1)$voltage)
33 sd(filter(voltage_data, location == 1)$voltage)
34 range(filter(voltage_data, location == 1)$voltage)
35
36 # QQ plot
37 qqnorm(voltage_data$voltage[voltage_data$location == 0], main = "Remote 0")
38 qqnorm(voltage_data$voltage[voltage_data$location == 1], main = "Local 1")

```

B.

```

41 Remote <- voltage_data$voltage_data[voltage_data$location == "0"]
42
43 Local <- voltage_data$voltage_data[voltage_data$location == "1"]
44 |
45 t_result <- t.test(Local, Remote,
46                   alternative = "two.sided",
47                   conf.level = 0.95,
48                   var.equal = FALSE)
49
50 print(t_result)
51
52 cat("95% CI for (Local - Remote):\n")
53 print(t_result$conf.int)
54
55 # Mean difference
56 cat("Mean Difference (Local - Remote):",
57     mean(local) - mean(remote), "\n")

```

QUESTION 2

Section 1

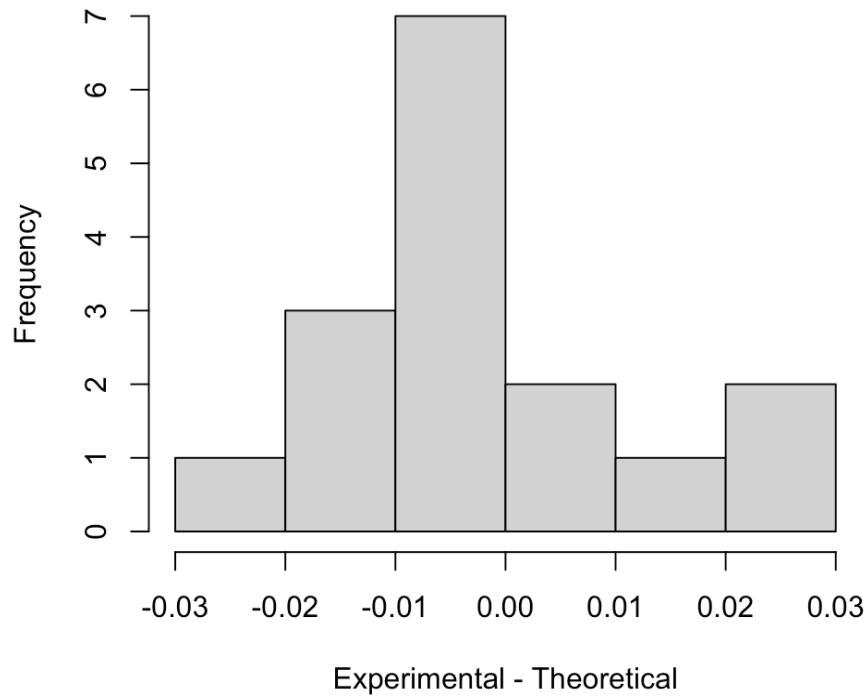
We define the problem as **difference(d) = experimental - theoretical**

H0 : mean(d) = 0 and Ha : mean(d) != 0

Since sample size is small, we also check for normality. We choose a paired t-test and analyze the difference.

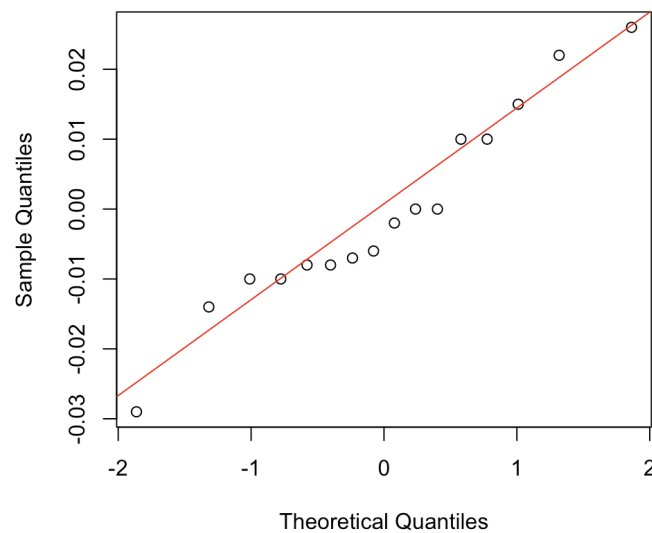
- The histogram for difference, shows that it is roughly symmetric and centers near 0.

Histogram of Differences



- The qq plot shows that the points lie close to the straight line, with only minor deviations.

Normal Q-Q Plot



- We can conclude that the distribution of the difference seems to be normal. Hence, we can use paired t-test for an analysis.

```
> print(t_test_result)
```

Paired t-test

```
data: vapor_data$experimental and vapor_data$theoretical
t = -0.19344, df = 15, p-value = 0.8492
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 -0.008262694  0.006887694
sample estimates:
mean difference
 -0.0006875
```

- The obtained p-value = 0.8492 >> 0.05, hence we fail to reject H0.
- And 0 also lies within the CI which gives us the same conclusion.
- There is no significant evidence that the mean difference between experimental and theoretical vapor pressure values is different from zero.

Values	
CI_lower	-0.00826269375004801
CI_upper	0.00688769375004801
margin_error	0.00757519375004801
mean_diff	-0.0006875000000000001
n	16L
sd_diff	0.014216041408681
t_crit	2.13144954555978

Section 2

Below code does the following:

- Captures difference between experimental and theoretical columns
- Summary statistics, histogram & boxplot
- QQ plot, paired t test
- Confidence Interval

```

1 vapor_data <- read.csv("/Users/fathimafaridamohammad/Downloads/vapor.csv")
2 # Capturing difference between experimental and theoretical data
3 vapor_data$diff <- vapor_data$experimental - vapor_data$theoretical
4 # Summary statistics
5 summary(vapor_data$diff)
6 # Histogram of differences
7 hist(vapor_data$diff, main = "Histogram of Differences",
8       xlab = "Experimental - Theoretical")
9 # QQ plot (check normality assumption)
10 qqnorm(vapor_data$diff)
11 qqline(vapor_data$diff, col = "red")
12 # Boxplot
13 boxplot(vapor_data$diff, main = "Boxplot of Differences")
14
15 ##### Paired t-test
16
17 t_test_result <- t.test(vapor_data$experimental,
18                         vapor_data$theoretical,
19                         paired = TRUE,
20                         alternative = "two.sided",
21                         conf.level = 0.95)
22
23 print(t_test_result)
24 ### Confidence Interval
25
26 mean_diff <- mean(vapor_data$diff)
27 sd_diff <- sd(vapor_data$diff)
28 n <- length(vapor_data$diff)
29 t_crit <- qt(0.975, df = n - 1)
30 margin_error <- t_crit * (sd_diff / sqrt(n))
31 CI_lower <- mean_diff - margin_error
32 CI_upper <- mean_diff + margin_error
33 cat("Mean Difference:", mean_diff, "\n")
34 cat("95% CI:", CI_lower, "to", CI_upper, "\n")

```

THE END